

# SS-31 (Elamipretide)

Mitochondrial-targeted tetrapeptide · 16 mg/mL injectable · Cellular bioenergetics

SubQ Injection

Once Daily

16 mg/mL

FDA Approved (Barth)

## COMMON INDICATIONS

- ▶ Barth syndrome (FDA-approved)
- ▶ Primary mitochondrial myopathy
- ▶ Mitochondrial dysfunction
- ▶ Heart failure adjunct
- ▶ Long COVID / post-viral fatigue
- ▶ Age-related cellular decline
- ▶ Chronic fatigue
- ▶ Neurodegenerative adjunct
- ▶ Athletic recovery / longevity

## STANDARD DOSING PROTOCOL (16 MG/ML)

|                                                      |                                                         |                                                         |                                                          |
|------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| <p>WEEK 1</p> <p><b>1–2 mg</b></p> <p>Daily SubQ</p> | <p>WEEKS 2–4</p> <p><b>3–5 mg</b></p> <p>Daily SubQ</p> | <p>STANDARD</p> <p><b>5–10 mg</b></p> <p>Daily SubQ</p> | <p>CYCLE</p> <p><b>8–12 wks</b></p> <p>Then reassess</p> |
|------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|

### DOSING NOTES

SS-31 selectively binds cardiolipin on the inner mitochondrial membrane, preserving cristae structure and ATP production. Inject SubQ daily, ideally in the morning (mitochondrial biogenesis peaks during active hours). Most clinical trials in heart failure and mitochondrial myopathy used 40 mg/day, but lower doses (5–10 mg) are typical in compounding clinical practice for general mitochondrial support. Doses  $\geq 15$  mg should be split across two injection sites. Reassess at 8–12 weeks; long-term safety data beyond 6 months are limited.

## VOLUME REFERENCE (16 MG/ML)

| TARGET DOSE   | VOLUME TO DRAW | U-100 SYRINGE | NOTES                                  |
|---------------|----------------|---------------|----------------------------------------|
| 1 mg          | 0.063 mL       | ~6 units      | Starting dose, sensitive patients      |
| 2 mg          | 0.125 mL       | ~13 units     | Initial titration                      |
| 5 mg          | 0.31 mL        | ~31 units     | Standard maintenance                   |
| 10 mg         | 0.625 mL       | ~63 units     | Higher dose, daily                     |
| 15 mg (split) | 2 × 0.47 mL    | 2 × 47 units  | Split across 2 sites, $\geq 2$ " apart |

## STORAGE & COUNSELING

### STORAGE

- ▶ Refrigerate at 36–46°F (2–8°C)
- ▶ Protect from light
- ▶ Do not freeze; do not shake vigorously
- ▶ BUD per pharmacy label

### CAUTIONS & COUNSELING

- ▶ Most common: mild injection site reaction
- ▶ Mild GI upset possible at higher doses
- ▶ Limited long-term safety data >6 months
- ▶ Pregnancy/lactation — not recommended